

UMM AL MARADIM, Kuwait

WMO: 405960

Lat: 28.680N

Lon: 48.652E

Elev: 8

StdP: 101.23

Time Zone: 3.00 (E03)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.9	12.3	-1.2	3.4	12.8	0.8	4.0	14.1	25.2	18.0	22.3	15.9	9.6	310	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.1	39.8	27.4	38.4	27.0	37.5	27.0	32.0	35.0	31.5	34.7	31.0	34.6	4.6	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.3	29.5	34.3	30.7	28.4	34.2	30.1	27.4	34.1	109.7	35.1	106.6	34.9	103.8	34.5	33.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.7	16.3	13.6	DB	9.0	42.3	1.5	1.8	7.9	43.6	7.0	44.7	6.2	45.7	5.1	47.1	(4)
(5)			WB	5.2	32.4	1.5	0.8	4.1	33.0	3.2	33.5	2.4	34.0	1.3	34.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.6	16.3	17.0	19.9	23.8	29.3	33.8	35.1	35.5	34.3	30.5	23.8	18.8		
	DBStd	7.34	2.38	1.95	1.82	1.80	2.30	1.77	1.33	1.22	1.17	1.79	2.67	2.25		
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	139	67	43	6	0	0	0	0	0	0	0	0	22		
	CDD10.0	6049	196	196	308	413	598	715	779	790	728	637	414	273		
	CDD18.3	3146	5	6	56	163	340	465	521	532	478	378	166	37		
	CDH23.3	43076	0	1	37	602	4134	7154	8423	8742	7659	5290	1026	7		
	CDH26.7	27092	0	0	3	58	1828	4754	5943	6262	5259	2841	144	0		
(14) Wind	WSAvg	6.2	7.6	6.8	6.2	5.0	5.7	6.2	5.8	5.4	5.8	6.0	7.2	7.2		
(15) Precipitation	PrecAvg	182	35	23	31	19	5	1	1	1	0	4	32	30		
	PrecMax	298	103	81	97	52	20	2	3	2	1	26	182	90		
	PrecMin	82	1	1	1	1	0	0	0	0	0	0	1	1		
	PrecStd	61	26	20	24	15	5	1	1	0	0	6	39	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.7	21.2	26.2	30.1	38.6	42.0	42.3	41.6	38.3	35.6	29.8	24.2		
		MCWB	17.5	16.9	19.0	20.3	27.4	30.0	28.8	28.7	26.5	26.4	22.6	19.7		
	2%	DB	20.2	20.4	24.2	28.2	34.9	39.2	39.5	39.3	37.2	34.4	29.1	23.4		
		MCWB	17.0	16.7	18.7	20.9	24.4	26.7	26.6	27.8	26.5	25.9	22.5	19.1		
	5%	DB	19.8	19.9	23.0	27.0	33.2	37.4	38.1	38.2	36.6	33.6	28.4	22.6		
		MCWB	16.5	16.6	18.3	20.8	24.2	26.0	26.6	27.7	26.5	25.6	21.7	18.2		
	10%	DB	19.3	19.4	22.2	26.1	32.1	36.2	37.2	37.3	36.0	32.9	27.5	21.9		
		MCWB	16.0	16.2	18.0	20.7	24.2	25.8	26.9	27.9	26.5	25.4	20.8	17.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.9	18.3	21.0	23.4	28.8	31.6	32.2	33.0	32.1	29.8	25.2	21.2		
		MCDB	20.3	20.0	24.0	27.4	35.8	38.8	35.1	36.0	34.6	33.4	28.5	23.3		
	2%	WB	18.1	17.9	19.7	22.6	27.2	30.1	31.5	32.2	30.9	28.6	24.1	20.2		
		MCDB	19.6	19.6	22.5	26.3	31.7	35.1	34.7	35.0	34.5	32.4	28.1	22.7		
	5%	WB	17.4	17.4	19.2	22.0	26.5	29.1	31.0	31.8	30.2	27.7	22.9	19.4		
		MCDB	19.4	19.3	21.8	25.7	30.8	34.3	34.5	34.6	34.1	32.1	27.4	22.1		
	10%	WB	16.7	16.9	18.8	21.5	25.8	28.2	30.4	31.4	29.6	27.0	21.8	18.1		
		MCDB	19.1	19.0	21.3	25.1	30.2	33.6	34.4	34.5	33.9	31.5	26.5	21.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	2.6	2.8	3.2	3.8	4.5	5.4	4.8	4.1	3.5	3.1	2.8	2.7		
		MCDBR	2.6	3.0	4.3	5.3	6.7	7.5	6.2	5.7	4.1	3.5	3.3	3.0		
		MCWBR	3.3	2.5	2.6	3.1	4.4	4.4	4.2	4.5	4.6	4.1	3.5	3.4		
	5% WB	MCDBR	2.4	2.5	3.2	4.4	5.9	6.6	5.0	4.3	3.3	3.2	2.8	2.7		
		MCWBR	3.6	2.4	2.5	3.0	4.4	4.9	4.1	3.7	4.2	4.2	3.8	3.7		
(40) Clear-Sky Solar Irradiance	taub		0.513	0.560	0.613	0.698	0.694	0.652	0.676	0.626	0.568	0.559	0.544	0.508		
	taud		1.874	1.779	1.664	1.565	1.577	1.632	1.622	1.717	1.819	1.832	1.851	1.898		
	Ebn at Noon		696	703	697	657	663	688	670	699	725	694	662	677		
	Edh at Noon		172	205	243	277	274	258	260	235	206	191	174	161		
(44) All-Sky Solar Radiation	RadAvg		3.35	4.11	5.16	5.65	6.53	7.20	6.85	6.57	5.93	4.78	3.49	3.07		
	RadStd		0.23	0.28	0.28	0.33	0.28	0.28	0.19	0.15	0.21	0.20	0.27	0.29		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	+0.33	N/A	N/A	+0.54	N/A	N/A	N/A	N/A	+94	+107

Nomenclature: See separate page